Gaussian Fermions

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I. FERMIONIC VECTOR SPACES

Fermions are conventionally defined in terms of creation and annihilation operators. Here we will instead start from fermionic vector spaces or state spaces and the properties which make them "fermionic." In the next section, we will see that the more familiar properties of creation and annihilation operators follow from the properties of the states.

The simplest fermionic vector space V has the basis

$$\mathcal{B}(V) = \{|0\rangle, |1\rangle\} \tag{1}$$

which is the basis for a single "spinless fermion," with $|0\rangle$ being the vacuum or empty state and $|1\rangle$ being the filled state or state of one fermion. So far the fermionic nature is not very obvious except that we can only have at most one fermion.

Many-body spaces can be formed by taking Cartesian products of spaces (tensor products of basis vectors) so we could consider a space V_1 with basis $\{|0\rangle_1, |1\rangle_1\}$ and space V_2 with basis $\{|0\rangle_2, |1\rangle_2\}$, then the space $V_1 \times V_2$ has a basis

$$\mathcal{B}(V_1 \times V_2) = \{|0\rangle_1 \otimes |0\rangle_2,\ |1\rangle_1 \otimes |0\rangle_2,\ |0\rangle_1 \otimes |1\rangle_2,\ |1\rangle_1 \otimes |1\rangle_2\}\ . \tag{2}$$

A generic vector (or "state") in this space has the form $|\psi\rangle = a\,|0\rangle_1 \otimes |0\rangle_2 + b\,|0\rangle_1 \otimes |1\rangle_2 + c\,|1\rangle_1 \otimes |0\rangle_2 + d\,|1\rangle_1 \otimes |1\rangle_2$.

The defining property of fermions is how one identifies vectors in the space $V_1 \times V_2$ with vectors in the space $V_2 \times V_1$. We can define this identification, or vector space isomorphism, by defining it on the basis vectors. It is given by:

$$\begin{aligned}
|0\rangle_1 \otimes |0\rangle_2 &\leftrightarrow +\,|0\rangle_2 \otimes |0\rangle_1 \\
|1\rangle_1 \otimes |0\rangle_2 &\leftrightarrow +\,|0\rangle_2 \otimes |1\rangle_1 \\
|0\rangle_1 \otimes |1\rangle_2 &\leftrightarrow +\,|1\rangle_2 \otimes |0\rangle_1 \\
|1\rangle_1 \otimes |1\rangle_2 &\leftrightarrow -\,|1\rangle_2 \otimes |1\rangle_1 \quad .
\end{aligned} \tag{3}$$

Note the minus sign in the last line. Plus signs are shown for emphasis.

We can summarize the vector space isomorphism as

$$|n_1\rangle_1 \otimes |n_2\rangle_2 \ \leftrightarrow\ (-1)^{n_1 \cdot n_2}\,|n_2\rangle_2 \otimes |n_1\rangle_1 \tag{4}$$

where $n_1, n_2 \in \{0, 1\}$. The factor in front of this isomorphism rule means the *ordering* of fermionic basis vectors matters: if two odd basis vectors are permuted, the basis vector changes by a minus sign. [**?**]

Note that tensor products of basis states such as $|1\rangle_i \otimes |1\rangle_j$ are commonly written in the more compact notation $|1\rangle_i\,|1\rangle_j$ or even $|1_i 1_j\rangle$. Tensor product symbols "$\otimes$" are generally dropped when understood from context, and will be dropped frequently in the rest of these notes.

II. FERMIONIC OPERATORS

A. Operator Definitions

For the simplest case of a "spinless fermion" space V with basis $\mathcal{B}(V) = \{|0\rangle, |1\rangle\}$, we can define the following operators:

$$\hat{n} = |1\rangle\langle 1| \tag{5}$$

$$\hat{c} = |0\rangle\langle 1| \tag{6}$$

$$\hat{c}^\dagger = |1\rangle\langle 0| \quad . \tag{7}$$

Together with the identity operator, this set defines a complete operator basis.

The $\hat{n}$ operator is known as the "number operator" since it "detects" the filling of a state:

$$\begin{aligned} \hat{n}\,|0\rangle &= 0 \cdot |0\rangle \\ \hat{n}\,|1\rangle &= 1 \cdot |1\rangle \quad . \end{aligned} \tag{8}$$

The $\hat{c}$ and $\hat{c}^\dagger$ are known as the "annihilation" and "creation" operators, respectively. This is simply because they map a filled state to an empty state or vice versa:

$$\hat{c}\,|1\rangle = |0\rangle \tag{9}$$

$$\hat{c}^\dagger\,|0\rangle = |1\rangle \quad . \tag{10}$$

One can of course define tensor products of operators in the usual way, such as $\hat{c}_1\hat{c}_2$ or $\hat{c}_2^\dagger\hat{c}_1$. [**?**] These operators act on the spaces $V_1 \times V_2$ or $V_2 \times V_1$, respectively.

B. Commutation Relations

Now comes a crucial point: the creation and annihilation operators have non-trivial commutation relations, and these are entirely a consequence of the state properties. For operators on state spaces V_i and V_j (with $i \neq j$), the commutation relations are

$$\hat{c}_i\hat{c}_j = -\hat{c}_j\hat{c}_i \tag{11}$$

$$\hat{c}_i^\dagger\hat{c}_j^\dagger = -\hat{c}_j^\dagger\hat{c}_i^\dagger \tag{12}$$

$$\hat{c}_i\hat{c}_j^\dagger = -\hat{c}_j^\dagger\hat{c}_i \tag{13}$$

The first two of these are commonly summarized as:

$$\begin{aligned} \{\hat{c}_i, \hat{c}_j\} &= 0 \\ \{\hat{c}_i^\dagger, \hat{c}_j^\dagger\} &= 0 \end{aligned} \tag{14}$$

where $\{a, b\} = ab + ba$ is the anti-commutator.

The third relation Eq. (13) has the special case that when $i = j$, $(\hat{c}_i\hat{c}_i^\dagger + \hat{c}_i^\dagger\hat{c}_i) = 1$. Therefore it is commonly summarized as

$$\{\hat{c}_i, \hat{c}_j^\dagger\} = \delta_{ij} \quad . \tag{15}$$

In many presentations, the anticommutation relations of the operators are stated as axioms, but they actually follow from the properties of the state space. For example,

$$\begin{aligned} \hat{c}_1\hat{c}_2 &= |0\rangle_1 \otimes \langle 1|_1 \otimes |0\rangle_2 \otimes \langle 1|_2 \\ &= |0\rangle_1 \otimes (\langle 1|_1 \otimes \langle 1|_2) \otimes |0\rangle_2 \\ &= |0\rangle_1 \otimes (-\langle 1|_2 \otimes \langle 1|_1) \otimes |0\rangle_2 \\ &= -|0\rangle_2 \otimes \langle 1|_2 \otimes \langle 0|_1 \otimes |1\rangle_1 \\ &= -\hat{c}_2 \otimes \hat{c}_1 \quad . \end{aligned} \tag{16}$$

C. Comment on State and Operator Labels

An important point for what follows is that the subscripts or labels on vectors such as $|0\rangle_i$ or $|1\rangle_j$ or on operators such as $\hat{c}_i$ or $\hat{c}_j$ index over spatial "sites" as well as spins, if present. For example, if the fermions are "spinless fermions" with sites labeled by integers then the labels are

$$i = 1, 2, 3, 4, \ldots \tag{17}$$

If the fermions are spinless and organized on a two-dimensional lattice then the labels could be

$$i = (1,1), (1,2), (1,3), \ldots, (2,1), (2,2), \ldots \tag{18}$$

Importantly, if the fermions are "spinful"—electrons being an important example—then the labels are understood to index both sites and spin values

$$i = (1,\uparrow),(1,\downarrow),(2,\uparrow),(2,\downarrow),\ldots \tag{19}$$

When spin plays an especially important role, such as in computing magnetization or magnetic susceptibility, then it can be helpful to explicitly display spin labels such as $\hat{c}_{i,\sigma}$ with $\sigma =\uparrow,\downarrow$.

The key point is that derivations below not explicitly showing spin labels or not explicitly referring to dimensionality are general and apply equally to spinless and spinful fermions in one-, two-, and three-dimensional space etc. with an appropriate understanding of the state and operator labels.

III. MANY-BODY STATES AND GAUSSIAN STATES

Basis states for many-body or many-particle systems can be defined or constructed as

$$|1_i 1_j 1_k\rangle = \hat{c}_i^\dagger \hat{c}_j^\dagger \hat{c}_k^\dagger |0\rangle \ . \tag{20}$$

The most general many-particle state is a sum of these:

$$|\Psi\rangle = \sum_{n_1,n_2,\ldots,n_N} \Psi^{n_1 n_2 \cdots n_N} |n_1 n_2 \cdots n_N\rangle \tag{21}$$

where $\Psi^{n_1 n_2 \cdots n_N}$ is an arbitrary tensor of complex amplitudes. The anti-commuting nature of the fermions is automatically satisfied by the properties of the basis, and no additional symmetry property needs to be assumed for the amplitudes $\Psi^{n_1 n_2 \cdots n_N}$.

A typical extra assumption, though, is that for a closed system the total number of particles N_p is fixed. Mathematically this means $n_1 + n_2 + \ldots + n_N = N_p$ otherwise $\Psi^{n_1 n_2 \cdots n_N} = 0$. (Superconducting states are a common exception to this rule, though these are more properly understood in terms of density matrices or mixed states.)

A special subset of many-particle fermionic states are the *Gaussian states*, more commonly known as *Slater determinants*. The general form of a Gaussian state with four fermions is:

$$\begin{aligned} |\Psi\rangle &= \hat{A}^\dagger \hat{B}^\dagger \hat{C}^\dagger \hat{D}^\dagger |0\rangle \\ &= \left(\sum_{i_1=1}^N A^{i_1} \hat{c}_{i_1}^\dagger\right)\left(\sum_{i_2=1}^N B^{i_2} \hat{c}_{i_2}^\dagger\right)\left(\sum_{i_3=1}^N C^{i_3} \hat{c}_{i_3}^\dagger\right)\left(\sum_{i_4=1}^N D^{i_4} \hat{c}_{i_4}^\dagger\right)|0\rangle \quad . \end{aligned} \tag{22}$$

Above we have introduced operators such as $\hat{A} = \sum_{i_1=1}^N A^{i_1} \hat{c}_{i_1}^\dagger$. One describes the state above by saying that it contains four fermions which are in the "orbitals" A, B, C and D. These orbitals are vectors $\vec{A} = [A^1\ A^2\ \ldots A^N]$ which are unit-normalized. The vectors are not required to be orthogonal, though the properties of the state are simpler to deduce if they are.

The Gaussian states are quite special and far from being a general fermionic state, even though they are elements of the general vector space of fermionic many-body states. To see that the Gaussian states are not general, let us write a two-particle Gaussian state $|AB\rangle = \hat{A}^\dagger \hat{B}^\dagger |0\rangle$ in the general form Eq. (21) for a fermionic state. We see that for the state $|AB\rangle$, the amplitudes $\Psi^{n_1 n_2 \cdots n_N}$ take the form:

$$\Psi^{0_1 0_2 \cdots 1_i \cdots 1_j \cdots 0_N} = A_i B_j \quad \text{for } \sum_i n_i = 2 \tag{23}$$

$$\Psi^{n_1 n_2 \cdots n_N} = 0 \qquad \text{otherwise for } \sum_i n_i \neq 2 \ . \tag{24}$$

So the amplitude tensor Ψ of $|AB\rangle$ is constructed from two vectors (it is rank-2 in a certain sense [**?**]) and is not fully general.

To understand the origin of the name Slater determinant, consider the basis defined as $|i\rangle = \hat{c}_i^\dagger |0\rangle$, the so-called "real-space basis". Expressing states in this basis is known as *first quantization*. We can express the amplitudes of $|AB\rangle$ in this basis, as $\psi_{AB}(i,j) = \langle i,j|AB\rangle = \langle 0| c_i^\dagger c_j^\dagger |AB\rangle$. One finds that

$$\psi_{AB}(i,j) = A^i B^j - B^i A^j \tag{25}$$

such that the amplitudes in real space take the form of an antisymmetric function. It was pointed out by Slater that expressions of the above type can be written as determinants of matrices whose columns are the orbital vectors $\vec{A}, \vec{B}, \ldots$ and whose rows are the spatial positions $i = 1, 2, \ldots, N$, hence the name Slater determinant.

The first-quantized basis can be convenient for proving facts such as if $\vec{A} = \vec{B}$, then the state $|AB\rangle = 0$, as is evident from Eq. (25) above. In the more general setting, the requirement that orbitals required be independent is known as the *Pauli exclusion principle*. It is not a separate axiom about fermions, but follows from their vector space properties.

IV. STATIC PROPERTIES OF GAUSSIAN STATES

In this section we will consider static or time-independent properties of Gaussian states. Here we consider only pure states. To define a generic Gaussian state, start by defining a matrix of orbitals $d^j{}_m$. This is assumed to be a unitary matrix $\bar{d}^{m'}{}_j d^j{}_m = \delta^{m'}_m$. The mth column $[\vec{d}_{(m)}]^j = d^j{}_m$ can be interpreted as the real-space wavefunction for the single fermion that will be created by the $\hat{d}^\dagger_m$ operator defined below.

Now define the operator $\hat{d}^\dagger_m$ and its conjugate as

$$\hat{d}^\dagger_m = \sum_j d^j{}_m \, \hat{c}^\dagger_j \tag{26}$$

$$\hat{d}^m = \sum_j \bar{d}^m{}_j \, \hat{c}^j \ . \tag{27}$$

From the unitary property of $g^j{}_m$ it follows that

$$\hat{c}^\dagger_j = \sum_m \bar{d}^m{}_j \, \hat{d}^\dagger_m \tag{28}$$

$$\hat{c}^j = \sum_m d^j{}_m \, \hat{d}^m \ . \tag{29}$$

The transformation from $\hat{c}$ to $\hat{d}$ operators above is a *canonical transformation*, which just means that the $\hat{d}$ operators obey the same commutation relations and have the same algebraic properties as the $\hat{c}$ operators, such as $\{\hat{d}^m, \hat{d}^\dagger_{m'}\} = \delta^m_{m'}$.

Furthermore, although we took the perspective above that the states built from the vacuum $|0\rangle$ by the $\hat{c}^\dagger_j$ operators are our defining, or "computational", basis, note the similarity of the transformations (26), (27) to the transformations (28), (29). One can just as equally view the states built up from the $\hat{d}^\dagger_m$ operators as the defining basis and conclude that properties like expected values of $\hat{c}$ or $\hat{c}^\dagger$ operators in such states will be equivalent up to replacing $d^j{}_m \leftrightarrow \bar{d}^m{}_j$.

A. Generic Gaussian State

The most generic form of a pure Gaussian state or Slater determinant is given by

$$|\mathbf{m}\rangle_d = |m_1 m_2 \ldots m_N\rangle_d = \hat{d}^\dagger_{m_1} \hat{d}^\dagger_{m_2} \cdots \hat{d}^\dagger_{m_N} |0\rangle \tag{30}$$

$$= \left[\prod_{m=1} (\hat{d}^\dagger_m)^{\eta_m} \right] |0\rangle \tag{31}$$

for some set of orbitals $d^j{}_m$ where η_m is the *occupancy* of state m in the d basis. The occupancy is defined here as $\eta_m = 1$ if $m \in \{m_1, m_2, \ldots, m_n\}$ and $\eta_m = 0$ otherwise. The total number of fermions is N.

B. Correlation matrix

A central quantity for Gaussian states is the correlation matrix $C^j_i = \langle \mathbf{m}|_d \, \hat{c}^\dagger_i \hat{c}^j \, |\mathbf{m}\rangle_d$. We will see that knowledge of the correlation matrix is sufficient to compute expected values of operators, the action of operators on Gaussian states, and dynamical properties such as single-particle Green's functions [**?**].

The correlation matrix of a Gaussian state $|\mathbf{m}\rangle_d$ is given by

$$\boxed{C_i^j = \langle \mathbf{m}|_d \, \hat{c}_i^\dagger \hat{c}^j \, |\mathbf{m}\rangle_d = \sum_m \bar{d}_i^m \eta_m d_m^j} \quad . \tag{32}$$

where recall that the d_m^j are the orbitals and η_m the occupancies defining the state.

To derive this result, note that for a Gaussian state $|\mathbf{n}\rangle_c = \left[\prod_{n=1} (\hat{c}_n^\dagger)^{\gamma_n}\right] |0\rangle$ the correlation matrix is $\langle \mathbf{n}|_c \, \hat{c}_i^\dagger \hat{c}^j \, |\mathbf{n}\rangle_c = \delta_i^j \gamma_i$, i.e. it is a diagonal matrix of the occupations. This result follows from the fact that $\hat{c}^j \, |\mathbf{n}\rangle_c = \gamma_j \, |\mathbf{n}'\rangle_c$ where $\mathbf{n}'$ is the set $\mathbf{n}$ with the element j removed if present. Then recalling that the transformation from the $\hat{c}$ operators to the $\hat{d}$ operators is a canonical transformation, a similar result holds in the d basis, namely $\langle \mathbf{m}|_d \, \hat{d}_m^\dagger \hat{d}^{m'} \, |\mathbf{m}\rangle_d = \delta_m^{m'} \eta_m$. Finally to derive Eq. (32) use (28) and (29) to write

$$C_i^j = \langle \mathbf{m}|_d \, \hat{c}_i^\dagger \hat{c}^j \, |\mathbf{m}\rangle_d = \sum_{m,m'} \bar{d}^m{}_i d^j{}_{m'} \, \langle \mathbf{m}|_d \, \hat{d}_m^\dagger \hat{d}^{m'} \, |\mathbf{m}\rangle_d = \sum_m \bar{d}_i^m \eta_m d_m^j \quad . \quad \square \tag{33}$$

For a normalized state $|\mathbf{m}\rangle_d$, note that the trace of the correlation matrix gives the number of fermions in the state

$$N \stackrel{\text{def}}{=} \sum_j \langle \mathbf{m}|_d \, \hat{n}_j \, |\mathbf{m}\rangle_d = \sum_j \langle \mathbf{m}|_d \, \hat{c}_j^\dagger \hat{c}^j \, |\mathbf{m}\rangle_d = \sum_j C_j^j = \sum_m \eta_m \ . \tag{34}$$

If the state is pure, the occupations η_m are either 0 or 1, so that the correlation matrix is a projector $C^2 = C$.

C. Expected Value of Quadratic Operators

Given a quadratic operator $\hat{\mathcal{O}} = \sum_{i,j} O^i{}_j \hat{c}_i^\dagger c^j$, we can use the correlation matrix result above to derive the expected value $\langle \mathbf{m}|_d \, \hat{\mathcal{O}} \, |\mathbf{m}\rangle_d$. It follows by linearity that

$$\boxed{\langle \mathbf{m}|_d \, \hat{\mathcal{O}} \, |\mathbf{m}\rangle_d = \sum_{i,j} O^i{}_j \, \langle \mathbf{m}|_d \, \hat{c}_i^\dagger c^j \, |\mathbf{m}\rangle_d = \sum_{i,j} O^i{}_j C_i^j = \mathrm{Tr}[OC]} \quad . \tag{35}$$

An important example of a quadratic operator is a non-interacting Hamiltonian, which in the particle-conserving case has the generic form $\hat{H} = \sum_{i,j} h^i{}_j \hat{c}_i^\dagger \hat{c}^j$. So from the result above we see that the energy expectation value of a Gaussian state depends only on its correlation matrix.

D. Action of Operators on Gaussian States

Given an arbitrary creation operator $\hat{v}^\dagger = v^j \hat{c}_i^\dagger$, where v^j is some vector of coefficients, consider its action on a Gaussian state $|\phi\rangle$, resulting in the state $|\tilde{\psi}\rangle = \hat{v}^\dagger \, |\phi\rangle$. The state $|\tilde{\psi}\rangle$ is a Gaussian state and its correlation matrix $(C_{\tilde{\psi}})_i{}^j = \langle \tilde{\psi}| \, \hat{c}_i^\dagger \hat{c}^j \, |\tilde{\psi}\rangle$ is given by

$$\boxed{C_{\tilde{\psi}} = ||v_0||^2 C_\phi + v_0 v_0^\dagger} \tag{36}$$

where C_ϕ is the correlation matrix of $|\phi\rangle$ and $v_0 = (1 - C_\phi)v$ is the projection of v onto the unoccupied subspace of C_ϕ.

Analogously, given an arbitrary annihilation operator $\hat{w} = \bar{w}_j \hat{c}^i$, where w^j is some vector of coefficients, consider its action on a Gaussian state $|\phi\rangle$, resulting in the state $|\tilde{\eta}\rangle = \hat{w} \, |\phi\rangle$. The state $|\tilde{\eta}\rangle$ is a Gaussian state and its correlation matrix $(C_{\tilde{\eta}})_i{}^j = \langle \tilde{\eta}| \, \hat{c}_i^\dagger \hat{c}^j \, |\tilde{\eta}\rangle$ is given by

$$\boxed{C_{\tilde{\eta}} = ||w_1||^2 C_\phi - w_1 w_1^\dagger} \tag{37}$$

where C_ϕ is the correlation matrix of $|\phi\rangle$ and $w_1 = C_\phi w$ is the projection of w onto the occupied subspace of C_ϕ.

Note that the states $|\tilde{\psi}\rangle$ or $|\tilde{\eta}\rangle$ are not normalized. This results in the trace of their correlation matrices $C_{\tilde{\psi}}$ or $C_{\tilde{\eta}}$ no longer having a trace equal to the correct particle number. (For $|\tilde{\psi}\rangle$ the correct particle number should be $N_\phi + 1$ and for $|\tilde{\eta}\rangle$ the correct particle number should be $N_\phi - 1$.) If desired, the correct normalization can be restored by multiplying by the appropriate factor.

E. Inner Product of Gaussian States

Consider two Gaussian pure states $|\mathbf{m}\rangle_d$ and $|\mathbf{n}\rangle_f$ given by:

$$|\mathbf{m}\rangle_d = |m_1 m_2 \ldots m_N\rangle_d = \hat{d}^\dagger_{m_1}\hat{d}^\dagger_{m_2}\cdots\hat{d}^\dagger_{m_N}|0\rangle \tag{38}$$

$$|\mathbf{n}\rangle_f = |n_1 n_2 \ldots n_N\rangle_f = \hat{f}^\dagger_{n_1}\hat{f}^\dagger_{n_2}\cdots\hat{f}^\dagger_{n_N}|0\rangle \tag{39}$$

$$\tag{40}$$

where the number of particles N is taken to be the same. (Otherwise the inner product of these states is zero.)

V. DYNAMICS OF GAUSSIAN STATES

An appealing aspect of Gaussian states is that their dynamical or time-dependent properties can be completely understood through simple and efficiently- computable matrix expressions. In this section we will consider dynamics governed by a quadratic Hamiltonian of the form

$$\hat{H} = \sum_{ij} h^i{}_j \hat{c}^\dagger_i \hat{c}^j \quad . \tag{41}$$

We will construct its eigenoperators and eigenstates then discuss how generic Gaussian states evolve under this Hamiltonian and show expressions for the Green's functions (time-dependent correlation functions) of these states.

A. Eigenoperators

To construct eigenoperators, begin by diagonalizing the Hamiltonian "hopping" or coefficient matrix $h^i{}_j$ as

$$h^i{}_j = \phi^i{}_n \epsilon_n \bar{\phi}_j{}^n \quad . \tag{42}$$

The resulting "single-particle eigenstates" $\phi^i{}_n$ will play a crucial role in what follows. These can be viewed as spatial single-particle wavefunctions or orbitals labeled by n.

Next construct eigenoperators defined as

$$\hat{\phi}^\dagger_n = \sum_j \phi^j{}_n \hat{c}^\dagger_j \tag{43}$$

$$\hat{\phi}^n = \sum_j \bar{\phi}_j{}^n \hat{c}^j \quad . \tag{44}$$

from which it follows by unitarity of ϕ^j_n that

$$\hat{c}^\dagger_j = \sum_n \bar{\phi}_j{}^n \hat{\phi}^\dagger_n \tag{45}$$

$$\hat{c}^j = \sum_n \phi^j{}_n \hat{\phi}^n \quad . \tag{46}$$

The reason we are calling these eigenoperators is that they only transform by a phase when evolving in time in the Heisenberg picture

$$\boxed{\hat{\phi}^n(t) = e^{-i\epsilon_n t}\ \hat{\phi}^n(0)} \tag{47}$$

$$\boxed{\hat{\phi}^\dagger_n(t) = e^{i\epsilon_n t}\ \hat{\phi}^\dagger_n(0)} \quad . \tag{48}$$

To show this result, use the fact that the $\hat{\phi}^\dagger_n$ and $\hat{\phi}^n$ obey the canonical fermionic anticommutation relations and use this to show $[\hat{\phi}^n, \hat{H}] = \epsilon_n \hat{\phi}^n$ and $[\hat{\phi}^\dagger_n, \hat{H}] = -\epsilon_n \hat{\phi}^\dagger_n$. These commutation relations hold for $\hat{\phi}^n(t)$ at all times. Then the properties (47) and (48) follow from the Heisenberg equation $i\frac{d}{dt}\hat{\phi}^n(t) = [\hat{\phi}^n(t), \hat{H}]$ and similarly for $\hat{\phi}^\dagger_n(t)$.

B. Eigenstates

Having found eigenoperators, we can construct many-body, Gaussian eigenstates. These are generically of the form

$$|\mathbf{n}\rangle_\phi = \prod_n \left(\hat{\phi}_n^\dagger\right)^{\nu_n} |0\rangle \tag{49}$$

where $\nu_n = 1$ if $n \in \{n_1, n_2, \ldots, n_N\}$ and $\nu_n = 0$ otherwise. Such states, known as *Fock states*, evolve only by a phase under Schrödinger time evolution

$$\boxed{e^{-i\hat{H}t} |\mathbf{n}\rangle_\phi = e^{-iE_\mathbf{n} t} |\mathbf{n}\rangle_\phi} \tag{50}$$

where

$$\boxed{E_\mathbf{n} = \epsilon_{n_1} + \epsilon_{n_2} + \ldots + \epsilon_{n_N}} \tag{51}$$

is the energy of the state.

To derive this result, we can use the Heisenberg picture to write

$$\begin{aligned} e^{-i\hat{H}t} |\mathbf{n}\rangle_\phi &= e^{-i\hat{H}t} \hat{\phi}_{n_1}^\dagger \hat{\phi}_{n_2}^\dagger \cdots \hat{\phi}_{n_N}^\dagger |0\rangle && (52) \\ &= \left(e^{-i\hat{H}t} \hat{\phi}_{n_1}^\dagger e^{i\hat{H}t}\right) \left(e^{-i\hat{H}t} \hat{\phi}_{n_2}^\dagger e^{i\hat{H}t}\right) \cdots \left(e^{-i\hat{H}t} \hat{\phi}_{n_N}^\dagger e^{i\hat{H}t}\right) |0\rangle && (53) \\ &= e^{-i(\epsilon_{n_1} + \epsilon_{n_2} + \ldots + \epsilon_{n_N})t} |\mathbf{n}\rangle_\phi && (54) \end{aligned}$$

where we have used $e^{-i\hat{H}t} \hat{\phi}_n^\dagger e^{i\hat{H}t} = \hat{\phi}_n(-t) = e^{-i\epsilon_n t} \hat{\phi}_n(0)$ and $e^{i\hat{H}t} |0\rangle = |0\rangle$.

Equation (51) is a crucial result, and justifies the level-filling picture used in band theory. It says we can understand the full, many-body spectrum of a Gaussian fermionic Hamiltonian just by enumerating the single-particle states ϕ_n^j for $n = 1, ..., V$ where V is the number of sites (volume) of the system. Then the 2^{N_S} different many-body eigenstates are just the exponentially many different ways of "filling" these N_S single particle states or levels.

C. Zero-Temperature Green's Functions

With these results in hand, it is straightforward to find expressions for the zero-temperature or ground state Green's functions. Their definitions are general for any many-body system while the expressions for them in terms of the ϕ_n^j matrix is specific to Gaussian states.

The lesser Green's function is given by

$$\boxed{G_j^{<\,i}(t) = i \langle\Phi_0| \hat{c}_j^\dagger \hat{c}^i(t) |\Phi_0\rangle = i \sum_n \nu_n \phi_n^i e^{-i\epsilon_n t} \bar{\phi}_j^n} \quad . \tag{55}$$

The greater Green's function is given by

$$\boxed{G_j^{>\,i}(t) = -i \langle\Phi_0| \hat{c}^i(t) \hat{c}_j^\dagger |\Phi_0\rangle = -i \sum_n (1 - \nu_n) \phi_n^i e^{-i\epsilon_n t} \bar{\phi}_j^n} \quad . \tag{56}$$

Finally the retarded Green's function is given by

$$\boxed{G_j^{R\,i}(t) = \theta(t)(G_j^{>\,i}(t) - G_j^{<\,i}(t)) = -i\theta(t) \sum_n \phi_n^i e^{-i\epsilon_n t} \bar{\phi}_j^n = -i\theta(t) e^{-iht}} \quad . \tag{57}$$

where $\theta(t) = 1$ for $t > 0$ and $\theta(t) = 0$ otherwise. In all the results above, the ν_n are the occupation numbers of the ground state of N fermions, which means they are $\nu_n = 1$ for the N lowest-energy single-particle eigenstates and $\nu_n = 0$ otherwise.

These results can all be derived using Eqs. (43) and (44) to transform the $\hat{c}$ and $\hat{c}^\dagger$ into linear combinations of the $\hat{\phi}$, $\hat{\phi}^\dagger$ operators. Then the results above follow from the time evolution of the $\hat{\phi}$ operators Eqs. (47) and (48).

The factor of $-i$ in the definition of G^R helps its Fourier transform have a particularly simple form. The Fourier transform of $f(t) = \theta(t)e^{-i\epsilon t}$ is $\tilde{f}(w) = i/(\omega - \epsilon + i\alpha)$ with $\alpha > 0$ a small broadening factor. Using this fact in Eq. (57) gives the expression for G^R in frequency space

$$\boxed{\tilde{G}^{R\,i}_j(\omega) = \sum_n \frac{\phi^i_n \bar{\phi}^n_j}{\omega - \epsilon_n + i\alpha} = \frac{1}{(\omega + i\alpha) - h}} \quad . \tag{58}$$

This form is known as the Lehmann representation. In the last expression above, the terms in the denominator are matrices and it is understood that $(\omega + i\alpha)$ is multiplied by an identity matrix.

D. Time Evolution of a Generic Gaussian State

Consider now a generic Gaussian state

$$|\mathbf{m}\rangle_d = |m_1 m_2 \dots m_N\rangle_d = \hat{d}^\dagger_{m_1} \hat{d}^\dagger_{m_2} \cdots \hat{d}^\dagger_{m_N} |0\rangle \tag{59}$$

$$= \left[\prod_{m=1} (\hat{d}^\dagger_m)^{\eta_m}\right] |0\rangle \quad . \tag{60}$$

where the $\hat{d}^\dagger_m$ operators are defined as

$$\hat{d}^\dagger_m = \sum_j d^j{}_m \hat{c}^\dagger_j \quad . \tag{61}$$

Define the *propagator* as $\boxed{g(t) = e^{-iht}}$ in other words $\boxed{g^i{}_j(t) = \sum_n \phi^i{}_n e^{-i\epsilon_n t} \bar{\phi}^n{}_j}$.

Using the fact that $\hat{d}^\dagger_m = \sum_j d^j_m \hat{c}^\dagger_j = \sum_{j,n} (\bar{\phi}^n_j d^j_m) \hat{\phi}^\dagger_n$, where we have used equations (61) and (45), it follows that

$$|\mathbf{m}\rangle_d = \sum_{j_1 j_2 \cdots} \sum_{n_1 n_2 \cdots} (\bar{\phi}^{n_1}_{j_1} d^{j_1}_{m_1})(\bar{\phi}^{n_2}_{j_2} d^{j_2}_{m_2}) \cdots (\bar{\phi}^{n_N}_{j_N} d^{j_N}_{m_N})\ \hat{\phi}^\dagger_{n_1} \hat{\phi}^\dagger_{n_2} \cdots \hat{\phi}^\dagger_{n_N} |0\rangle \tag{62}$$

which implies, using Eq. (54) that

$$e^{-i\hat{H}t} |\mathbf{m}\rangle_d = \sum_{j_1 j_2 \cdots} \sum_{n_1 n_2 \cdots} (\bar{\phi}^{n_1}_{j_1} d^{j_1}_{m_1})(\bar{\phi}^{n_2}_{j_2} d^{j_2}_{m_2}) \cdots (\bar{\phi}^{n_N}_{j_N} d^{j_N}_{m_N})\ e^{-i(\epsilon_{n_1} + \epsilon_{n_2} + \dots)t}\ \hat{\phi}^\dagger_{n_1} \hat{\phi}^\dagger_{n_2} \cdots \hat{\phi}^\dagger_{n_N} |0\rangle \tag{63}$$

$$= \sum_{j_1 j_2 \cdots} \sum_{n_1 n_2 \cdots} (\phi^{j'_1}_{n_1} e^{-i\epsilon_{n_1} t} \bar{\phi}^{n_1}_{j_1} d^{j_1}_{m_1})(\phi^{j'_2}_{n_2} e^{-i\epsilon_{n_2} t} \bar{\phi}^{n_2}_{j_2} d^{j_2}_{m_2}) \cdots (\phi^{j'_N}_{n_N} e^{-i\epsilon_{n_N} t} \bar{\phi}^{n_N}_{j_N} d^{j_N}_{m_N})\ \hat{c}^\dagger_{n_1} \hat{c}^\dagger_{n_2} \cdots \hat{c}^\dagger_{n_N} |0\rangle \tag{64}$$

$$= \sum_{j_1 j_2 \cdots} \sum_{n_1 n_2 \cdots} (g^{j'_1}{}_{j_1}(t) d^{j_1}_{m_1})(g^{j'_2}{}_{j_2}(t) d^{j_2}_{m_2}) \cdots (g^{j'_N}{}_{j_N}(t) d^{j_N}_{m_N})\ \hat{c}^\dagger_{n_1} \hat{c}^\dagger_{n_2} \cdots \hat{c}^\dagger_{n_N} |0\rangle \tag{65}$$

$$= \hat{d}^\dagger_{m_1}(t) \hat{d}^\dagger_{m_2}(t) \cdots \hat{d}^\dagger_{m_N}(t) |0\rangle \quad . \tag{66}$$

where we have defined

$$\boxed{\hat{d}^\dagger_m(t) = \sum_{j'} d^{j'}{}_m(t) \hat{c}^\dagger_{j'} = \sum_{j'j} \left(g^{j'}{}_j(t) d^j{}_m\right) \hat{c}^\dagger_{j'}} \quad . \tag{67}$$

We therefore see that generic orbitals, defined by a matrix of orbitals $d^j{}_m$, evolve in time by acting with the propagator matrix $g^{j'}{}_j(t)$ onto the orbital matrix $d^j{}_m$.

As a special case of this result, it is interesting to observe that if we choose the orbitals to be the eigenorbitals $d^j{}_m = \phi^j{}_m$ then we find $\phi^j{}_m(t) = g^{j'}{}_j(t)\phi^j{}_m = e^{-i\epsilon_m t}\phi^j{}_m$.

E. Time Evolution of a Generic Correlation Matrix

As we saw above, for a Gaussian state $|\mathbf{m}\rangle_d$ with correlation matrix $C^j_i = \langle \mathbf{m}|_d\, \hat{c}^\dagger_i \hat{c}^j\, |\mathbf{m}\rangle_d = \sum_m \bar{d}^m_j \eta_m d^j{}_m$, the orbitals evolve, in the Schrödinger picture, as $d^i{}_m(t) = g^i{}_j(t) d^j{}_m$. So we find that

$$\boxed{C^j_i(t) = \bar{g}^{i'}_i(t)\, C_{i'}{}^{j'}\, g^j_{j'}(t)} \quad . \tag{68}$$

VI. THERMAL GAUSSIAN STATES

The Gaussian formalism extends straightforwardly to finite temperature, also offering closed-form results for all properties of interest. Given a Gaussian Hamiltonian $\hat{H} = \sum_{i,j} h^i_j \hat{c}^\dagger_i \hat{c}^j$, the finite-temperature (mixed) state in the canonical ensemble formalism at temperature T or coolness $\beta = 1/T$ has the form

$$\hat{\rho} = \frac{1}{Z}\, e^{-\beta \hat{H}} \quad . \tag{69}$$

Given a spectral decomposition of the "hopping matrix" $h^i_j = \sum_n \phi^i_n \epsilon_n \hat{\phi}^n_j$ and defining $\hat{\phi}^\dagger_n = \sum_j \phi^j_n \hat{c}^\dagger_j$, we can then write

$$\hat{\rho} = \frac{1}{Z}\, e^{-\beta \sum_n \epsilon_n \hat{\phi}^\dagger_n \hat{\phi}^n} \quad . \tag{70}$$

VII. CORRELATION MATRIX

The correlation matrix of a thermal Gaussian state is

$$C_i{}^j = \mathrm{Tr}[\hat{c}^\dagger_i \hat{c}^j] = \left[\frac{1}{1+e^{\beta h}}\right]^j_i \tag{71}$$

VIII. ENTANGLEMENT AND SCHMIDT DECOMPOSITION OF GAUSSIAN STATES

Consider a pure Gaussian state $|\psi\rangle$ with correlation matrix $C_i{}^j = \langle\psi|\, \hat{c}^\dagger_i \hat{c}^j\, |\psi\rangle$. Our goal is to obtain a Schmidt decomposition across two disjoint regions A and B meaning

$$|\psi\rangle = \sum_\mu w_\mu\, |\mu\rangle_A\, |\mu\rangle_B \tag{72}$$

where $|\mu\rangle_A$ is an orthonormal basis of A and $|\mu\rangle_B$ an orthonormal basis of B. To find this decomposition, it will turn out to be sufficient to diagonalize the block of C restricted to A, called C_{AA}. Below we will find an expression for the Schmidt values or weights w_μ in terms of the eigenvalues of C_{AA}.

A. Properties of Correlation Matrix Blocks

First we will establish some useful relationships between blocks of a correlation matrix. The only assumptions needed will be that the correlation matrix of a pure Gaussian state is a projector and is Hermitian:

$$C^2 = C \tag{73}$$

$$C^\dagger = C \quad . \tag{74}$$

Consider the blocks of C formed by restricting its rows and columns to regions A or B (permuting the rows and columns if needed):

$$C = \begin{bmatrix} C_{AA} & C_{AB} \\ C_{BA} & C_{BB} \end{bmatrix} \tag{75}$$

Because $C^\dagger = C$, it follows that $C^\dagger_{AB} = C_{BA}$ and that C_{AA} and C_{BB} are Hermitian. Then the property $C^2 = C$ implies the following relations

$$C_{AB}C_{BA} = C_{AA}(1 - C_{AA}) \tag{76}$$

$$C_{BA}C_{AB} = C_{BB}(1 - C_{BB}) \tag{77}$$

$$C_{AA}C_{AB} = C_{AB}(1 - C_{BB}) \tag{78}$$

$$C_{BB}C_{BA} = C_{BA}(1 - C_{AA}) \quad . \tag{79}$$

B. Block Decomposition

Without loss of generality, take the linear size m of the space B to be greater than or equal to linear size n of A, so $m \leq n$. Then C_{AA} is an $n \times n$ matrix, C_{BB} is $m \times m$, and C_{AB} is $n \times m$.

To obtain the Schmidt decomposition, first compute the singular value decomposition of C_{AB}

$$C_{AB} = A\Lambda B^\dagger \quad . \tag{80}$$

where $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots)$ is a square, real, non-negative, diagonal matrix of linear dimension r where $r = \min(m, n)$. The matrices A and B are $n \times r$ and $m \times r$, respectively, and obey $B^\dagger B = 1$ and $A^\dagger A = 1$.

Note that $C_{AB}C_{BA} = A\Lambda^2 A^\dagger$. By using Eq. (76) we can obtain the relation:

$$C_{AA}(1 - C_{AA}) = A\Lambda^2 A^\dagger \quad . \tag{81}$$

Thus C_{AA} is also diagonalized by the matrix A

$$C_{AA} = A\Omega A^\dagger \tag{82}$$

where Ω is a diagonal matrix $\Omega = \mathrm{diag}(\nu_1, \nu_2, \ldots)$ of "occupations" (r of these). Equation (81) implies the eigenvalues ν_j of C_{AA} are related to the singular values of C_{AB} through

$$\lambda_j = \sqrt{\nu_j(1 - \nu_j)} \quad . \tag{83}$$

Since C is a projector, one can show that the eigenvalues of diagonal blocks such as C_{AA} lie between 0 and 1. Each ν_j is within the range $0 \leq \nu_j \leq 1$ so the singular values of C_{AB} are in the range $0 \leq \lambda_j \leq 1/2$.

To complete the analysis of the blocks, we likewise find that

$$C_{BB}(1 - C_{BB}) = B\Lambda^2 B^\dagger \tag{84}$$

To make the above equation consistent with the decompositions of C_{AB} and C_{AA} above, we can conclude that C_{BB}, which we took to be as large or larger than C_{AA} has two kinds of eigenvalues:

1. Eigenvalues $\tilde{\nu}_j = (1 - \nu_j)$ which fall between $0 \leq \tilde{\nu}_j < 1$, where recall ν_j is an eigenvalue of C_{AA}.

2. Eigenvalues $\tilde{\nu}_j = 1$.

One can understand the second case of eigenvectors of C_{BB} which have eigenvalue $\tilde{\nu}_j = 1$ as spanning the space not captured by the column space of the rectangular matrix B. These $\tilde{\nu}_j = 1$ orbitals have spatial support which is entirely inside of region B.

C. Construction of Filled and Empty Orbitals